EN3160 Assignment 2: Fitting and Alignment

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GitHub: https://github.com

1 Sunflower blob detection using LoG

The image was read with `cv2.IMREAD_REDUCED_COLOR_4` (a 360×360 image). For each σ, the image was smoothed with a Gaussian and the scale-normalized response $\sigma^2|\nabla^2(G_\sigma * I)|$ was computed. Local extrema were retained across neighbouring scales and pixels. To focus detections on flowers, the sky/horizon and image border were excluded; radius was estimated using $r \approx \sqrt{2}\sigma$. The tested range was $\sigma = 1.0$ to 9.7 pixels in steps of 0.3 pixels.

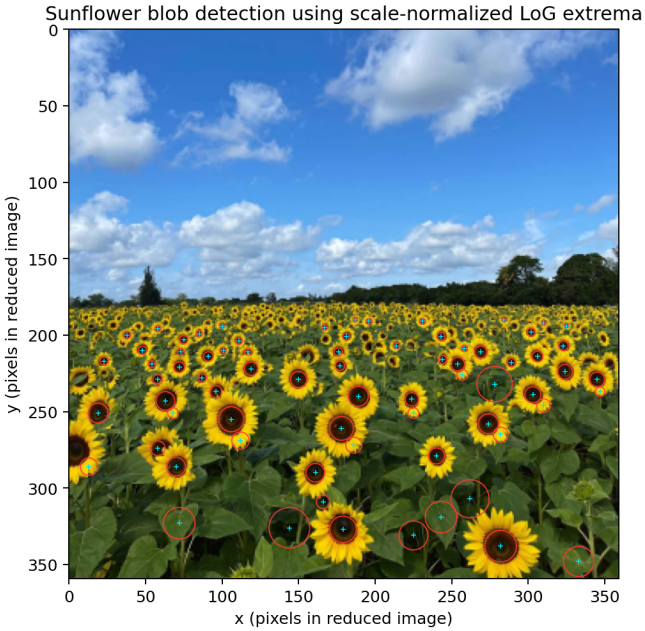


Figure 1: Detected LoG scale-space extrema shown as circles; coordinates and radii are measured in the reduced image.

Rank	Centre (x, y)	σ	Radius r (px)	Response
1	(144, 326)	9.4	13.29	0.666
2	(262, 307)	8.8	12.45	0.526
3	(278, 232)	8.2	11.60	0.550
4	(282, 338)	7.3	10.32	1.388
5	(72, 323)	7.3	10.32	0.275
6	(243, 319)	7.3	10.32	0.273

The detector returns approximate blob scales rather than exact petal boundaries; nearby flowers, leaves, and strong edges can affect the LoG response. The largest accepted blobs are reported above, sorted by estimated radius. **Core LoG code:**

```
R = []
for sigma in sigmas:
    smooth = cv2.GaussianBlur(gray, (0, 0), sigmaX=float(sigma))
    R.append(sigma**2 * np.abs(cv2.Laplacian(smooth, cv2.CV_32F)))
scale_space = np.stack(R)
peaks = scale_space == maximum_filter(scale_space, size=(3, 7, 7))
```

2 Line and circle fitting with RANSAC

The noisy point set was regenerated from Listing 1 with random seed 0 for repeatability. My line RANSAC samples two points, uses the unit normal (a, b) and the perpendicular distance $|ax + by - d|$, and selects the largest consensus. The line was then refined using total least squares (TLS). After removing the line consensus, circle RANSAC samples three points, rejects nearly collinear samples, and tests radial error $|\sqrt{(x - x_c)^2 + (y - y_c)^2} - r|$. The circle was refined by minimizing the sum of squared radial residuals over its RANSAC inliers. Thresholds were 1.4 for the line and 1.2 for the circle.

2.1 RANSAC results and discussion

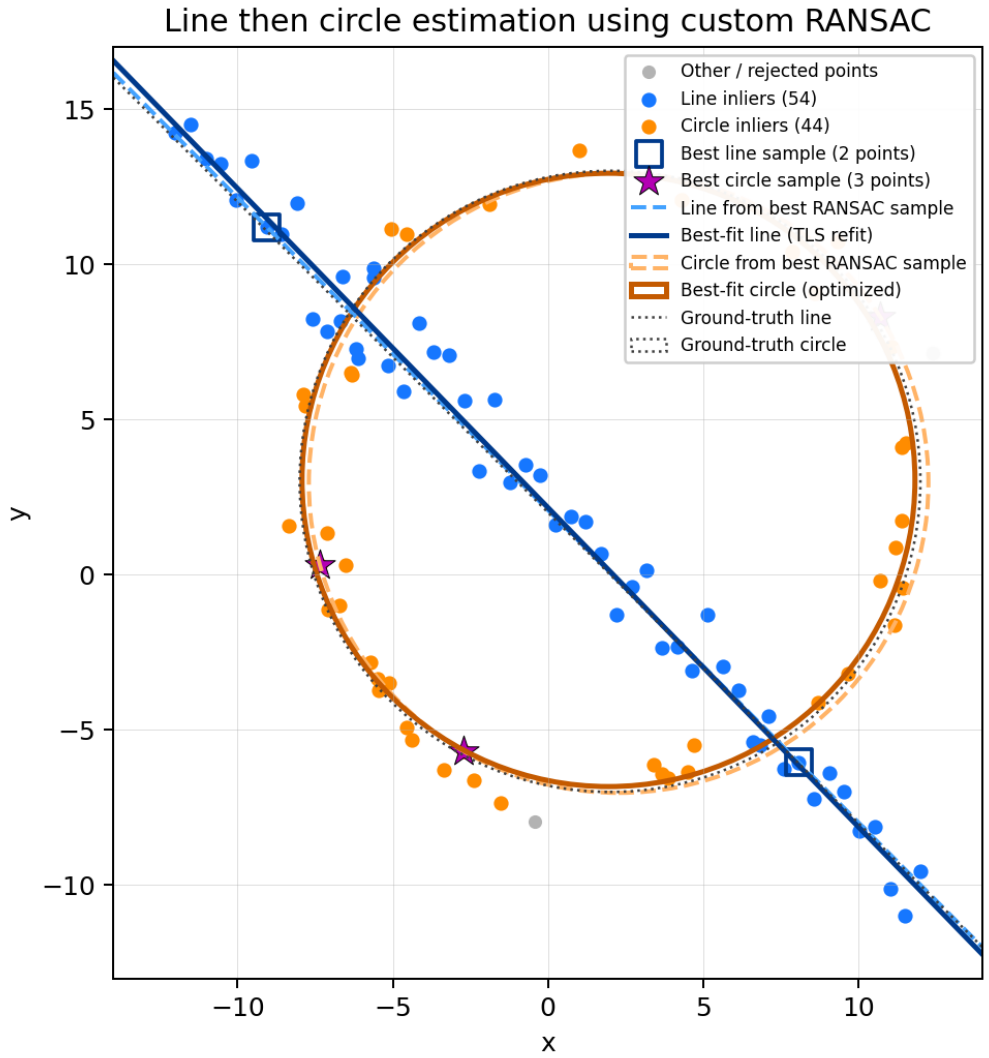


Figure 2: The best sample models (dashed), refined models (solid), inliers, sampled points, and known generating models (dotted).

Model	Estimated parameters	Consensus
Line	$(a, b, d) = (0.7167,\ 0.6974,\ 1.5029)$	54 points
Unit-normal check	$a^2 + b^2 = 1.0000$	threshold $= 1.4$
Circle	$(x_c, y_c, r) = (1.9512,\ 3.0461,\ 9.8732)$	44 points
Circle threshold	radial error < 1.2	sample size $= 3$

The fitted line is $0.7167x + 0.6974y = 1.5029$, or approximately $y = -1.0276x + 2.1546$, close to the generating line $y = -x + 2$. The fitted circle centre is about 0.067 units from $(2, 3)$ and its radius error is about 0.127 units. Some circle points fall within the line threshold and are removed with the line consensus, which explains why the circle has fewer than 50 inliers.

What if the circle were fitted first? With suitable thresholds and rejection of degenerate triples, circle-first can also work. However, three nearly collinear points from the line define a very large-radius circle; such samples can be numerically unstable and may incorrectly collect line points as a circle consensus. Fitting the line first removes most of that competing structure before the circle stage. **Core RANSAC inlier tests:**

```
model = line_from_two_points(X[i], X[j])  # unit-normal line from 2 points
inliers = line_distances(X, model) < tau_line
radial_error = np.abs(np.linalg.norm(X - centre, axis=1) - radius)
circle_inliers = radial_error < tau_circle
```

3 Homography and planar compositing

A homography was estimated from four ordered source corners to four destination corners on a planar facade. The warped texture was blended inside the destination quadrilateral. The UK flag is a familiar visual texture for the right-hand facade; a self-designed computer-vision poster was used as a second texture on the left-hand facade. The coordinates are editable in `solution.py`; for a different photograph, choose four corners in clockwise order (top-left, top-right, bottom-right, bottom-left) and update the points.

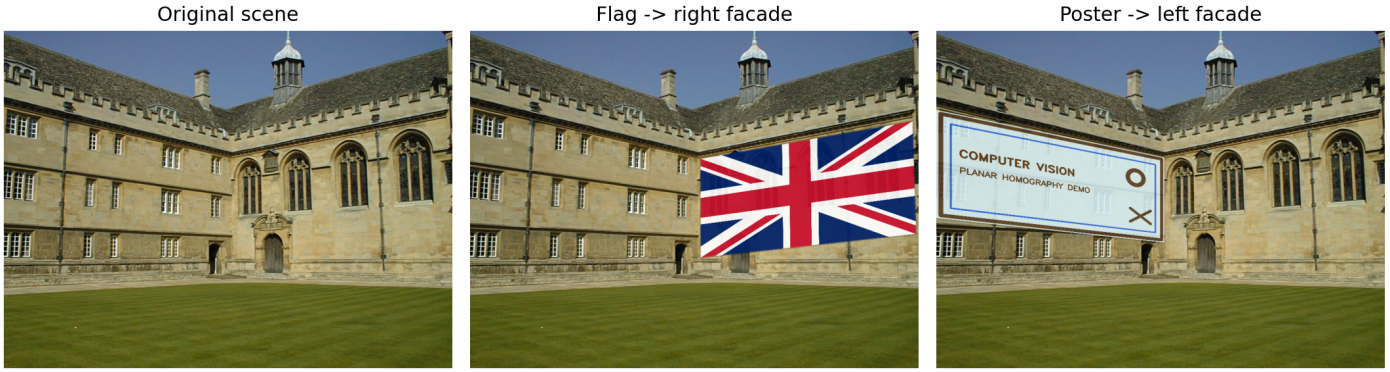


Figure 3: Original Wadham College scene and two homography-based composites.

In homogeneous coordinates, the mapping is $\lambda[x', y', 1]^T = H[u, v, 1]^T$, where H is estimated from the four source/destination correspondences. The texture is warped with `cv2.warpPerspective`; a warped mask restricts alpha blending to the selected quadrilateral. The destination points (top-left, top-right, bottom-right, bottom-left) are: right facade $[(526, 291), (1010, 205), (1018, 462), (526, 516)]$; left facade $[(0, 179), (520, 287), (520, 482), (0, 426)]$.

The transform corrects perspective for the chosen plane, but the overlay does not automatically reason about occlusion. Windows and architectural details below the inserted texture can be covered; a more realistic result would need a carefully chosen surface mask and manual occlusion handling. If a second, different architectural photograph is required by the marker, replace the scene and set four destination points for its own planar surface.

4 PCB alignment and difference images

`AECC1.jpg` was used as the template. SIFT keypoints/descriptors were detected, candidate descriptors were matched to the template with `cv2.BFMatcher`, Lowe's ratio test was applied, and a homography was estimated with RANSAC. Each candidate was warped into the template dimensions, then an absolute grayscale difference was computed and evaluated within a mask around the template board. Difference figures are contrast-scaled for visibility; the unscaled masked grayscale difference is also saved in the output folder. **Core homography and alignment code:**

```
H = cv2.getPerspectiveTransform(src_quad, dst_quad)  # Q3: four corners
texture_warp = cv2.warpPerspective(texture, H, (scene_w, scene_h))
sift = cv2.SIFT_create()
pairs = cv2.BFMatcher(cv2.NORM_L2).knnMatch(desc_img, desc_ref, k=2)
good = [m for m, n in pairs if m.distance < ratio * n.distance]
H, inliers = cv2.findHomography(src_pts, ref_pts, cv2.RANSAC, 5.0)  # Q4
aligned = cv2.warpPerspective(candidate, H, (ref_w, ref_h))
difference = cv2.absdiff(gray_ref, cv2.cvtColor(aligned, cv2.COLOR_BGR2GRAY))
```

4.1 PCB matching and differences

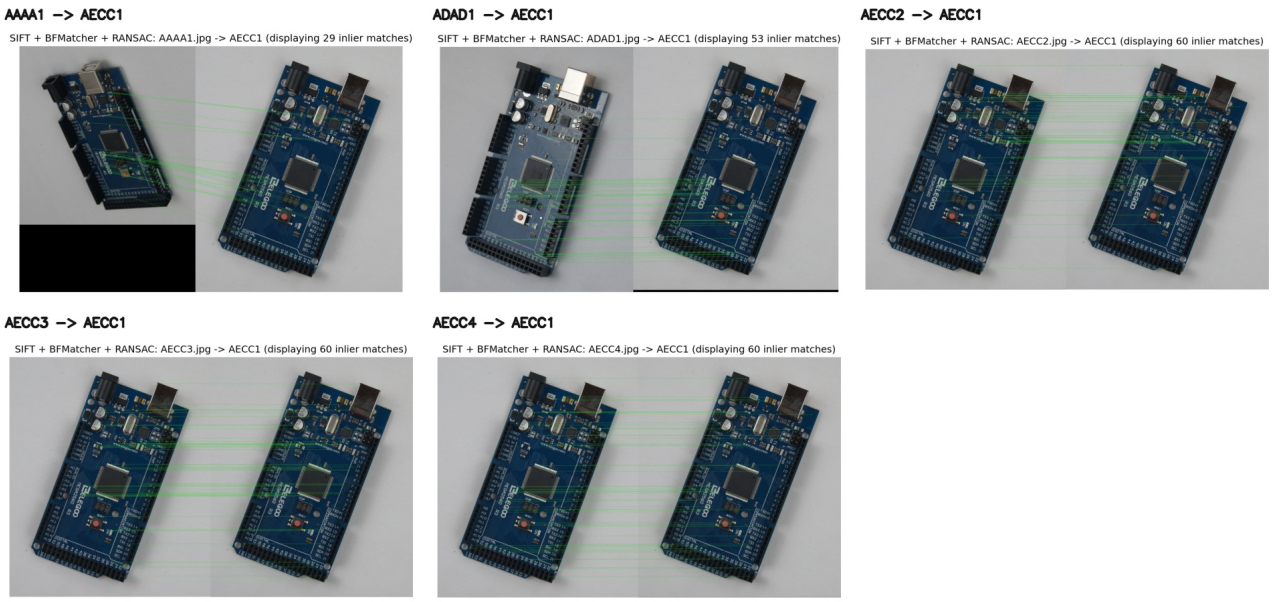


Figure 4: SIFT feature matches that are classified as homography inliers by RANSAC. All five non-template PCB images are included.

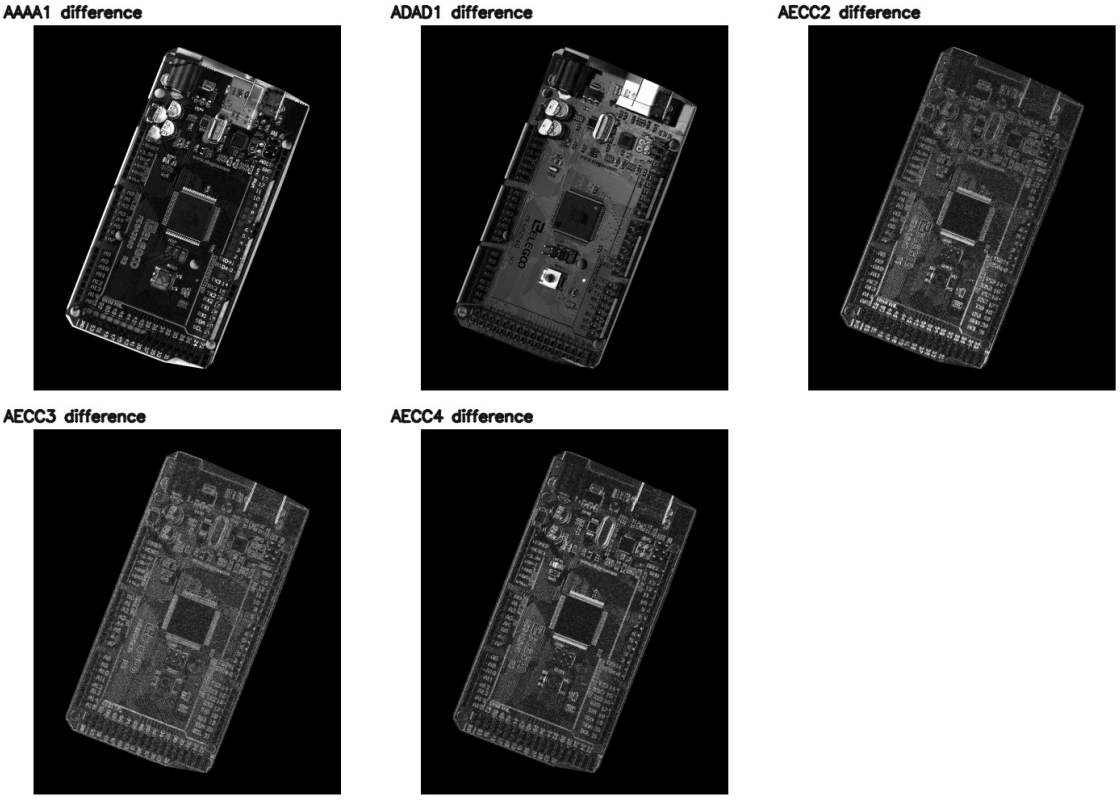


Figure 5: Normalized grayscale absolute-difference images after warping each PCB image to `AECC1.jpg`.

Candidate	Good matches	RANSAC inliers	Mean board difference
AAAA1	183	29	26.71
ADAD1	263	53	45.88
AECC2	1062	905	4.59
AECC3	1287	1130	3.96
AECC4	910	778	4.97

The AECC2–AECC4 photographs align closely with the template and have lower mean board differences. AAAA1 and ADAD1 have fewer consistent matches and larger differences, reflecting stronger viewpoint/appearance changes and likely board/photo variation. The USB Type-B connector, capacitors, and reset button can show high local differences because they protrude above the PCB plane. A single homography models one plane; it cannot simultaneously align elevated components, their shadows, and the board surface under perspective changes. Lighting, reflections, and small translations also contribute.